**NAME: Anmol** **ROLL NO: U24EV052**

# EXPERIMENT NO. : 9

**DATE: 16/10/2026**

## 1. AIM

To deposit Bismuth Oxyiodide (BiOI) thin film using Thermal Chemical Vapor Deposition (t-CVD) by transporting $BiI_3$ precursor vapors through a quartz tube under $Ar/O_2$ flow.

## 2. APPARATUS REQUIRED

The following apparatus were used for the experiment:

- Three-zone tube furnace
- Quartz tube
- Ceramic crucible
- $BiI_3$ precursor powder (approximately 500 mg)
- Argon (Ar) gas as carrier gas
- Oxygen ($O_2$) gas as oxidant
- Microscope glass slide (substrate)
- Silver paste for fixing the substrate
- Mass Flow Controllers (MFCs)

## 3. THEORY

Chemical Vapor Deposition (CVD):

Chemical Vapor Deposition (CVD) is a thin-film deposition technique in which volatile precursor materials are transported in the vapor phase to a heated substrate, where chemical reactions take place and form a thin film.

In thermal CVD (t-CVD), the precursor is vaporized by heating. The vapor is transported through the quartz tube using a carrier gas and reacts with the supplied reactant gas to form the required material on the substrate.

BiOI Formation:

Bismuth Oxyiodide (BiOI) is formed when vaporized $BiI_3$ reacts with oxygen. In this experiment, $BiI_3$ powder acts as the precursor, while oxygen acts as the oxidant. The reaction takes place in the reaction region of the quartz tube, resulting in the formation of a BiOI thin film:

$$2\ BiI_3\ (g) + O_2\ (g) \rightarrow 2\ BiOI\ (s) + 2\ I_2\ (g)$$

Role of Temperature Zones:

The three-zone tube furnace provides different temperature conditions for precursor vaporization, chemical reaction, and deposition:

- Zone 1 (Precursor Zone): Maintained at a higher temperature of 360 °C to vaporize the $BiI_3$ precursor.
- Zone 2 (Reaction Zone): Maintained at 150 °C, where the transported $BiI_3$ vapor reacts with oxygen to form BiOI. The substrate is positioned in this zone.
- Zone 3 (Downstream Zone): Maintained at room temperature or mild heating. Temperature gradients in this region influence the deposition of the material.

The temperature gradient along the quartz tube helps in controlling the vapor transport and deposition process. A ring-type deposition pattern can naturally form in the downstream region under suitable deposition conditions.

## 4. PROCEDURE

1. Preparation of Substrate

• Microscope glass slides were cleaned and dried.

• Substrates were attached to the glass slide using silver paste.

• The slide was loaded into the quartz tube at the downstream side.

• The substrate was positioned at a distance of 14.5 cm from the precursor, in Zone 2.

2. Loading of Precursor

• About 500 mg of $BiI_3$ powder was placed in a ceramic crucible.

• The crucible was placed in Zone 1 of the tube furnace.

3. System Setup and Gas Flow

• The quartz tube was assembled and sealed.

• A mixture of Ar (19 mL/min) and $O_2$ (4.5 mL/min) was set using MFCs.

• The gas flow was allowed to purge the tube.

4. Temperature Setting

The three temperature zones were set as follows:

• Zone 1 (precursor zone): 360 °C

• Zone 2 (reaction zone): 150 °C

• Zone 3 (downstream zone): Room temperature to mild heating.

5. Deposition Process

• The furnace was ramped to the set temperatures.

• $BiI_3$ was vaporized in Zone 1 and carried by $Ar/O_2$ through Zone 2.

• The vapor reacted with oxygen, forming BiOI.

• Deposition occurred naturally at the substrate position in Zone 2, forming a visible ring-type BiOI film.

6. Cooling

• After the process duration (typically 30–60 minutes), the furnace was turned off.

• The system was allowed to cool before handling the deposited substrate.

## 6. PARAMETERS

**Table 1: Parameters Used for BiOI Thin-Film Deposition**

| S.No. | Parameter | Value |
|---|---|---|
| 1 | $BiI_3$ Precursor Quantity | 500 mg |
| 2 | Zone 1 Temperature | 360 °C |
| 3 | Zone 2 Temperature | 150 °C |
| 4 | Zone 3 Temperature | Room temperature to mild heating |
| 5 | Argon Gas Flow Rate | 19 mL/min |
| 6 | Oxygen Gas Flow Rate | 4.5 mL/min |
| 7 | Substrate Position | 14.5 cm from precursor, in Zone 2 |
| 8 | Deposition Duration | 30–60 minutes |

## 7. OBSERVATIONS

During the experiment, $BiI_3$ precursor powder was heated in Zone 1 of the three-zone tube furnace, producing vapors that were transported through the quartz tube with the help of argon and oxygen gas. The vapor reacted with oxygen in Zone 2, resulting in the formation of Bismuth Oxyiodide (BiOI).

The substrate was positioned at a distance of 14.5 cm from the precursor in Zone 2, which was maintained at 150 °C. A visible ring-type deposition pattern of BiOI was observed on the substrate. The experiment demonstrated the effect of temperature zones, gas flow, and substrate position on the thin-film deposition process.

## 8. CONCLUSION

The experiment provided practical understanding of thermal Chemical Vapor Deposition (t-CVD) for the preparation of BiOI thin films. It demonstrated the importance of precursor vaporization, carrier and oxidant gas flow, and temperature control in the deposition process. The use of a three-zone furnace helped in understanding how different temperature regions contribute to precursor transport, chemical reaction, and thin-film formation. The experiment also provided insight into the formation of ring-type deposition and the influence of substrate positioning on the deposition process.

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# EXPERIMENT NO. : 10 (ASSIGNMENT 5)

**DATE: 16/10/2026**

## 1. AIM

To study the importance of spacers in FET fabrication and understand their formation, patterning, and role in advanced semiconductor device structures. To analyze spacer deposition and patterning in the Spacer and 14nm FinFET process flows using Semulator3D.

## 2. TOOL USED

Semulator3D

## 3. THEORY (SPACER)

Introduction to Spacer:

A spacer is a thin insulating layer formed on the sidewalls of the gate structure in MOSFET and FinFET devices. It is generally made of dielectric materials such as silicon nitride ($Si_3N_4$) or silicon dioxide ($SiO_2$).

Spacers are formed by depositing a dielectric layer over the gate structure, followed by an etching process that removes material from horizontal surfaces while retaining it along the vertical sidewalls.

Importance and Applications of Spacer:

Spacers play an important role in semiconductor device fabrication. Their major applications include:

• Source/Drain Formation: Help control the position and separation of source/drain regions from the gate.

• Implantation Mask: Act as a mask during ion implantation to control dopant placement.

• Short Channel Effect Control: Help in controlling source/drain extension dimensions in scaled transistors.

• Capacitance Reduction: Reduce unwanted parasitic capacitance between the gate and source/drain regions.

• Device Performance: Help maintain proper device dimensions and improve transistor performance.

Spacers are widely used in modern MOSFET and FinFET fabrication, particularly in advanced technology nodes where precise control over transistor dimensions is essential.

## 4. ASSIGNMENT QUESTION 1: PROCESS EDITOR STEPS

**Table: Steps involved in Spacer Formation (Process Editor Recipe)**

| Step | Action | Step Name | Material(s) | Depth | Mask | Polarity |
|---|---|---|---|---|---|---|
| 1 | Wafer Setup | Wafer Setup | /Silicon/Si_Xtal | 100 | – | – |
| 2 | Sequence | Fin Formation | – | – | – | – |
| 2.1 | Conformal Deposit | Silicon Nitride Hard Mask | /Nitrides/Si3N4 | 20 | – | – |
| 2.2 | Planarizing Deposit | Resist for SiN Hard Mask | /Organics/Resist | 20 | – | – |
| 2.3 | Expose Material | Expose Resist | /Organics/Resist | 20 | Fin_Pattern | Light |
| 2.4 | Enhanced Basic Etch | SiN Hard Mask Etch | /Nitrides/Si3N4 | 20 | – | – |
| 2.5 | Remove Material | Remove Resist | /Organics/Resist | – | – | – |
| 2.6 | Taper Etch | Etch for Fin Formation | /Silicon/ | 60 | – | – |
| 2.7 | Remove Material | Remove SiN Hard Mask | /Nitrides/Si3N4 | – | – | – |
| 3 | Sequence | STI | – | – | – | – |
| 3.1 | Planarizing Deposit | Planar Deposit | /Oxides/SiO2 | 0 | – | – |
| 3.2 | Enhanced Basic Etch | Basic Etch | /Oxides/SiO2 | 30 | – | – |
| 4 | Sequence | Gate Metal Patterning | – | – | – | – |
| 4.1 | Conformal Deposit | Gate High-K | /Oxides/HfOx | 2 | – | – |
| 4.2 | Conformal Deposit | Gate Metal Deposition | /Metals/TiN | 5 | – | – |
| 4.3 | Planarizing Deposit | Gate Metal Deposition | /Metals/W | 20 | – | – |
| 4.4 | Planarizing Deposit | Resist for Gate Patterning | /Organics/Resist | 100 | – | – |
| 4.5 | Expose Material | Expose Resist | /Organics/Resist | 0 | Gate | Light |
| 4.6 | Enhanced Basic Etch | Metal Etch | /Metals/TiN | 60 | – | – |
| 4.7 | Remove Material | Remove Resist | /Organics/Resist | – | – | – |
| 5 | Sequence | Spacer Pattern | – | – | – | – |
| 5.1 | Conformal Deposit | Spacer Deposition | /Nitrides/Si3N4 | spacer_thk | – | – |
| 5.2 | Enhanced Basic Etch | Spacer Etch | /Nitrides/Si3N4 | spacer_thk*1.1 | – | – |

## 4. ASSIGNMENT QUESTION 1 (A) & (B)

(a) Deposition Technique Used for Spacer:

The deposition technique used for spacer formation is Conformal Deposition. In this process, a thin layer of silicon nitride ($Si_3N_4$) is deposited uniformly over the gate structure, covering both horizontal and vertical surfaces.

Importance of Conformality:

Conformality is important because it ensures uniform thickness of the deposited layer over the entire structure, including the vertical sidewalls. This helps in obtaining properly shaped and uniform spacers on both sides of the gate.

Common Conformal Deposition Techniques:

• Chemical Vapour Deposition (CVD)

• Low Pressure Chemical Vapour Deposition (LPCVD)

• Atomic Layer Deposition (ALD)

• Plasma Enhanced Chemical Vapour Deposition (PECVD)

(b) Why is Spacer Patterned Without Lithography?

Spacer patterning is performed without lithography because it uses an anisotropic etching process. After conformal deposition, the deposited material is etched in the vertical direction. This removes the material from horizontal surfaces while retaining it on the vertical sidewalls of the gate.

Therefore, spacers are formed automatically on both sides of the gate without requiring an additional lithography mask. This process is known as self-aligned spacer formation.

## 4. ASSIGNMENT QUESTION 1 (C) & (D)

(c) Calculation of CD Difference at 50% Height:

The target CD is 7 nm, whereas the simulated CD width obtained from the process simulation is 7.01171875 nm. The CD at 50% of the total spacer thickness is 6.649137126980349 nm.

Total spacer thickness: $T_{spacer}$ = 53.83402490615845 nm

Height at 50% of the total spacer thickness: $H_{50\%} = (50 / 100) \times T_{spacer}$ = 26.9170124531 nm

CD difference is calculated as:

$\Delta CD = CD_{simulated} - CD_{50\%}$ = 7.01171875 nm - 6.649137126980349 nm = 0.362582 nm

(d) CD Profile Along the Spacer Height:

The CD profile represents the variation in critical dimension along the spacer height. The actual height at each position is calculated using the total spacer thickness: $H = (\text{Height (\%)} / 100) \times T_{spacer}$. The CD difference at each height is calculated as: $\Delta CD = CD_{simulated} - CD_{\text{at given height}}$.

**Table 1: CD Variation Along the Spacer Height**

| Sr. No. | Height (nm) | CD (nm) | ΔCD (nm) |
|---|---|---|---|
| 1 | 2.691701 | 6.972015 | 0.039704 |
| 2 | 5.383402 | 6.902577 | 0.109142 |
| 3 | 8.075104 | 6.745773 | 0.265946 |
| 4 | 10.766805 | 6.738281 | 0.273438 |
| 5 | 13.458506 | 6.732538 | 0.279180 |
| 6 | 16.150207 | 6.718109 | 0.293610 |
| 7 | 18.841909 | 6.703679 | 0.308040 |
| 8 | 21.533610 | 6.689249 | 0.322470 |
| 9 | 24.225311 | 6.673857 | 0.337861 |
| 10 | 26.917012 | 6.649137 | 0.362582 |
| 11 | 29.608714 | 6.625379 | 0.386340 |
| 12 | 32.300415 | 6.588619 | 0.423099 |
| 13 | 34.992116 | 6.546088 | 0.465631 |
| 14 | 37.683817 | 6.451959 | 0.559760 |
| 15 | 40.375519 | 6.376195 | 0.635524 |
| 16 | 43.067220 | 6.275944 | 0.735774 |
| 17 | 45.758921 | 6.132113 | 0.879606 |
| 18 | 48.450622 | 5.918727 | 1.092992 |
| 19 | 51.142324 | 4.983500 | 2.028218 |

## 5. QUESTION 2: PROCESS ANALYSIS OF 14 NM FINFET

(a) Spacer Deposition and Patterning Process Steps:

The following table shows the important process steps involved in spacer deposition and patterning in the 14 nm FinFET structure:

**Table 2: Spacer deposition and patterning process steps in 14 nm FinFET**

| Sr. No. | Process Step | Action Performed |
|---|---|---|
| 1 | 2.1.15 | Conformal deposition of silicon dioxide ($SiO_2$) to form the fin spacer oxide layer. |
| 2 | 2.1.16 | Etching of the deposited fin spacer oxide to retain the required spacer structure. |
| 3 | 2.6.1.1 | Deposition of silicon dioxide ($SiO_2$) for P-type spacer formation. |
| 4 | 2.6.1.2 | Deposition of silicon nitride ($Si_3N_4$) over the P-type spacer structure. |
| 5 | 2.6.1.5 | Etching of the deposited P-type spacer nitride layer. |
| 6 | 2.6.2.1 | Deposition of silicon nitride ($Si_3N_4$) for N-type spacer formation. |
| 7 | 2.6.2.4 | Etching of the deposited N-type spacer nitride layer. |

## 5. MATERIALS USED FOR SPACER

(b) Materials Used for Spacer:

In the 14 nm FinFET structure, two dielectric materials are used for spacer formation:

• Silicon Dioxide ($SiO_2$): Used for fin spacer oxide formation and P-type spacer formation.

• Silicon Nitride ($Si_3N_4$): Used for P-type and N-type spacer formation.

## 6. OBSERVATIONS

1. Spacer formation was observed on the vertical sidewalls of the gate structure after conformal deposition and anisotropic etching.

2. In Question 1, the simulated CD width was approximately 7.0117 nm. The CD gradually decreased along the spacer height, and the calculated CD loss at 50% of the spacer height was approximately 0.3626 nm.

3. In Question 2, the spacer deposition and patterning steps in the 14 nm FinFET process flow were identified.

4. Silicon dioxide ($SiO_2$) and silicon nitride ($Si_3N_4$) were identified as the dielectric materials used for spacer formation.

5. Separate spacer formation steps were observed for fin, P-type, and N-type regions.

## 7. CONCLUSION

The spacer formation process was studied using the Spacer.zam project and the inbuilt 14nmFinFET.zam example in Semulator3D. The importance of conformal deposition and anisotropic etching in forming self-aligned spacers was understood.

The CD variation along the spacer height was analyzed, and the CD loss at 50% of the spacer height was calculated. The deposition and patterning steps in the 14 nm FinFET process were also identified. The use of silicon dioxide and silicon nitride as spacer materials was studied. Overall, this assignment provided an understanding of spacer formation and its importance in MOSFET and FinFET fabrication.